

Q2-Commerce: Product and Sprint Backlog

Product Backlog

The Product Backlog contains all high-level features prioritized for development, ensuring the platform meets both business and technical requirements.

ID	Feature	Description
P-01	User Authentication	Implement JWT-based authentication for secure login and role management.
P-02	Product Management	CRUD operations for vendors to manage product inventory.
P-03	Shopping Cart	Users can add/remove items, track cart contents.
P-04	Transaction Processing	Enable checkout flow with Stripe and PayPal integration.
P-05	Order Tracking	Customers can view order history and track delivery status.
P-06	Messaging System	Real-time communication between customers and vendors.
P-07	User Reviews & Ratings	Customers can rate vendors/products based on purchases.
P-08	Admin Dashboard	Admins manage users, privileges, and transactions.
P-09	Theme Customization	Implement Dark, Grey, and Light themes.
P-10	Search & Filtering	Advanced product search with category-based filtering.
P-11	Mobile Responsiveness	Ensure adaptive UI for all screen sizes.
P-12	Security Enhancements	Firestore security rules and data encryption.
P-13	Performance Optimization	Implement caching and database indexing.
P-14	Scalability Preparations	Ensure horizontal scaling for high traffic load.

Sprint Backlogs

The development process is organized into multiple sprints, each with specific goals and tasks.

Sprint 1: Foundation Setup & Core Features

Sprint Goal

 Establish authentication, user management, and essential product handling features.

Sprint Duration: 2 Weeks

Task ID	Sprint Backlog Item	Description	Assigned To	Status
S-01	User Authentication	Implement JWT-based authentication and role handling	Backend Dev	In Progress
S-02	User Signup/Login	Create signup & login endpoints with validation	Backend Dev	To Do
S-03	Firebase Firestore Setup	Configure Firestore database structure	Backend Dev	Done
S-04	UI Login Page	Develop React components for login and signup	Frontend Dev	In Progress
S-05	Product CRUD	Implement endpoints for product creation, update, delete	Backend Dev	To Do
S-06	Product Card UI	Design ProductCard component in React	Frontend Dev	In Progress
S-07	Shopping Cart	Build Add to Cart and Remove Item functionality	Frontend Dev	To Do
S-08	Security Rules	Define Firestore rules for user access control	Backend Dev	To Do
S-09	Admin Role Management	Ensure administrators can manage users and privileges	Backend Dev	To Do

Sprint 2: Transactions, Messaging & Order Tracking

Sprint Goal



Develop checkout system, order tracking, and real-time messaging features.

Sprint Duration: 2 Weeks

Task ID	Sprint Backlog Item	Description	Assigned To	Status
S-10	Transaction Processing	Implement checkout flow with Stripe & PayPal	Backend Dev	To Do
S-11	Order History	Users can access completed orders in Firestore	Backend Dev	To Do
S-12	Order Tracking	Enable tracking functionality in UI	Frontend Dev	To Do
S-13	Messaging System	Integrate message component tied to transactions	Backend Dev	To Do
S-14	UI Messaging Component	Build React messaging interface for vendors/customers	Frontend Dev	To Do
S-15	User Reviews	Implement customer/vendor feedback and ratings	Backend Dev	To Do
S-16	Review Display UI	Design ReviewCard UI to display feedback	Frontend Dev	To Do
S-17	Performance Optimization	Implement caching and indexing for faster query results	Backend Dev	To Do
S-18	Theme Customization	Ensure all UI components support theme toggling	Frontend Dev	To Do

Sprint 3: Search, Filtering & UI Enhancements

Sprint Goal

✅ Enable advanced search functionality, improve mobile responsiveness, and refine UI for scalability.

Sprint Duration: 2 Weeks

Task ID	Sprint Backlog Item	Description	Assigned To	Status
S-19	Product Search	Build search functionality with category-based filtering	Backend Dev	To Do
S-20	Search UI	Create search bar and filter interface in React	Frontend Dev	To Do
S-21	Mobile Responsiveness	Ensure UI adapts seamlessly to different screen sizes	Frontend Dev	To Do
S-22	Performance Optimization	Use lazy loading for images and components	Frontend Dev	To Do
S-23	Scalability Preparations	Implement Firestore sharding and horizontal scaling	Backend Dev	To Do
S-24	Admin Dashboard	Refine dashboard components for managing user roles	Frontend Dev	To Do
S-25	Notification System	Integrate alerts for and messages	Backend Dev	To Do

Sprint 4: Security & Final Refinements

Sprint Goal

✅ Finalize security protocols, scalability enhancements, and optimize performance across the system.

Sprint Duration: 2 Weeks

Task ID	Sprint Backlog Item	Description	Assigned To	Status
S-26	Security Rules	Refine Firestore rules for enhanced security	Backend Dev	To Do
S-27	Data Encryption	Implement encryption for sensitive information	Backend Dev	To Do
S-28	Scalability Enhancements	Finalize horizontal scaling capabilities	Backend Dev	To Do
S-29	UI Refinements	Conduct final UI tests and adjustments for responsiveness	Frontend Dev	To Do
S-30	Stress Testing	Perform load tests to ensure performance under traffic spikes	QA Team	To Do
S-31	Final Bug Fixes	Address issues identified during QA testing	Backend/Frontend Devs	To Do

Sprint Schedule Summary

Sprint	Goal	Duration
Sprint 1	Authentication & Product Handling	2 weeks
Sprint 2	Transactions & Messaging	2 weeks
Sprint 3	Search & Filtering UI Enhancements	2 weeks
Sprint 4	Security & Optimization	2 weeks